

Acme Retail Inc.

E-Commerce Platform Migration to OCI

Architecture Proposal — March 2026



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Prepared with OCI Deal Accelerator

This document is provided for informational purposes only. Actual results may vary.

Executive Summary

Reduce Oracle Database licensing costs by 60% and modernize to cloud-native architecture

Current State

- 3 Oracle 19c databases on Exadata X8M on-prem (largest 4 TB OLTP)
- WebLogic 14c application servers (6x bare metal)
- GoldenGate replication to reporting database
- MQ Series for async messaging
- Current Oracle licensing: \$2M/year

Target State

Migrate to OCI Autonomous Database (ADB-S) with OKE for application tier, full HA/DR

Timeline: 6-month migration, Q3 2026 go-live

Service Tiering

Each tier drives HA/DR topology, backup strategy, isolation model, and support level.

Workload	Tier	Uptime	RTO	RPO
Main OLTP (4 TB)	Gold	99.99%	< 4 hrs	15 min
Reporting DB (2 TB)	Silver	99.5%	1 day	1 day
APEX App DB (200 GB)	Bronze	98%	Best effort	Best effort

Architecture Principles

Design

- **D-01 Operational Simplicity** — keep it simple, simplify again
- **D-02 Application First** — SaaS > PaaS > IaaS
- **D-05 HA for Business Continuity** — built-in, not bolted-on
- **D-06 Compliance by Design** — PCI baked in from day 1

Deployment

- **P-01 Automation First** — Terraform for all infrastructure
- **P-02 Environment Separation** — Prod / Pre-Prod / Dev isolated
- **P-04 Data Security** — TDE + Key Vault from day 1

Service

- **S-01 Cloud Monitoring** — OCI native tools (Monitoring, Logging, Ops Insights)
- **S-04 Cloud Onboarding** — PoC before full migration

Architecture Overview

[Insert diagram — export from examples/ecommerce-architecture.drawio] Tenancy → Region (Ashburn) → VCN with 3 subnets: • Public: Flexible LB + WAF + Bastion • App Tier: OKE Cluster (3 workers, auto-scaling) • Data Tier: ADB-S OLTP (4 OCPU, BYOL) + ADB-S DW + NoSQL + Redis DR Region (Phoenix): ADB-S Standby via Autonomous Data Guard Connectivity: FastConnect 1 Gbps (redundant) + VPN backup

Architecture Decisions

Decision	Rationale
ADB-S over ExaCS for primary database	Variable workload pattern, no dedicated DBA team, auto-scaling covers seasonal peaks. ExaCS oversized for this workload.
OKE over VM-based app tier	Team moving to containers. OKE provides auto-scaling per microservice, free control plane, blue/green deployments.
Cross-region ADG for DR	PCI compliance requires geo-redundant disaster recovery. ADG provides seconds RPO, minutes RTO.
FastConnect over VPN for hybrid connectivity	Low-latency requirement during migration coexistence. VPN as backup path only.
BYOL licensing for ADB-S	Customer has 8 existing processor licenses. BYOL saves ~75% on OCPU cost vs License Included.
OCI Streaming over dedicated MQ migration	Kafka-compatible API allows gradual migration from MQ Series. No new middleware licensing.

High Availability & Disaster Recovery

Active-passive DR with Autonomous Data Guard. Application tier provisioned on-demand in DR region via Terraform.

Component	Technology	RTO	RPO
Database (Primary)	ADB-S + TAC (Transparent Application Continuity)	~0 perceived	0 (local)

Component	Technology	RTO	RPO
Database (DR)	Cross-Region Autonomous Data Guard (async)	< 15 minutes	< 30 seconds
Application	OKE Auto-scaling + Health Checks	Seconds	N/A (stateless)
Application (DR)	Terraform on-demand provisioning	15-30 minutes	N/A (stateless)

Security & Compliance

✓ PCI-DSS | ✓ SOC2

Identity

- IAM least-privilege per compartment
- MFA for all human users
- Federation with Okta (SAML)
- Instance principals for service auth

Network

- Private subnets for app and data tiers
- NSGs (not security lists)
- WAF on all internet-facing endpoints
- Service Gateway (no internet traversal to OCI)

Database

- TDE with customer-managed keys (Vault)
- Data Safe (audit, masking, VPD)
- Private endpoints only
- Key rotation < 90 days

Monitoring

- Cloud Guard with detector recipes
- OCI Audit (365-day retention)
- VCN Flow Logs
- Vulnerability Scanning on OKE nodes

Environment Catalogue

Environment	Tier	Databases	OCPUs	Isolation
Production	Gold	OLTP (4 TB)	8	Dedicated CDB, private endpoint
Production	Silver	Reporting (2 TB)	4	Shared CDB OK
Production	Bronze	APEX (200 GB)	2	Shared CDB OK
Pre-Prod	Silver	OLTP copy	4	Refreshable clone from Prod
Dev/Test	Bronze	OLTP copy	2	Full clone, auto-stop nights/wknd
DR	Gold	OLTP standby	8	Cross-region ADG, dedicated

Cost Optimization

- Pre-Prod uses Refreshable Clones (read-only, no OCPU when paused)
- Dev/Test auto-stops nights and weekends (saves ~65%)
- DR standby billed at same OCPU rate as primary

Cost Estimate

Component	Monthly (PAYG)	Monthly (BYOL)	Notes
ADB-S Primary (4 OCPU, auto-scale to 12)	\$5,376	\$1,344	Auto-scale billed per OCPU-second
ADB-S Standby (ADG, same region)	\$5,376	\$1,344	Same shape as primary
ADB-S DR (Cross-region, Phoenix)	\$5,376	\$1,344	Async replication
ADB-S Storage (2 TB x3)	\$710	\$710	6 TB total across 3 instances
OKE Cluster (3x E4.Flex, 4 OCPU / 64 GB)	\$657	—	Control plane free
Networking (FastConnect 1 Gbps)	\$300	\$300	Redundant circuits
Storage & Other	\$450	\$450	Object Storage, Block, Vault
Total Monthly	\$18,245	\$5,492	
Total Annual	\$218,940	\$65,904	BYOL saves \$153K/year

Assumptions:

- Pricing based on US East (Ashburn) region, March 2026 list prices
- Auto-scaling costs estimated at 15% of base OCPU cost (seasonal peaks)
- Data transfer costs excluded (minimal with FastConnect)
- Oracle Support costs not included (existing contract assumed)

Migration Approach

Phase	Timeline	Key Activities
Phase 1: Assessment	Weeks 1-4	Run ADB assessment tool Inventory all DB objects Identify incompatibilities
Phase 2: Infrastructure	Weeks 3-6	Provision OCI tenancy Deploy landing zone Configure FastConnect
Phase 3: Migration	Weeks 5-10	ZDM for database migration Deploy OKE workloads Data validation
Phase 4: Cutover	Weeks 9-12	Final sync DNS cutover Post-migration validation Decommission on-prem

Migration Tools: OCI Zero Downtime Migration (ZDM), OCI Database Migration Service, Data Pump

Downtime Approach: Zero downtime via ZDM with GoldenGate-based replication

Operational Responsibilities (Co-Managed)

Activity	Customer	Oracle / Partner
DB provisioning	R	C
Patching & upgrades	R	C
Backup & recovery	R/A	C
Performance monitoring	R	I
Security configuration	R/A	C
Incident response (DB)	R	C
DR testing	R	C
Access control (IAM)	R/A	I
Schema changes	R/A	I
Oracle SR escalation	R	C

R = Responsible | A = Accountable | C = Consulted | I = Informed

Risk Register

Risk	Severity	Mitigation
Application compatibility with ADB-S restrictions	MEDIUM	Run ADB assessment tool pre-migration. Pilot with non-prod database first.

Risk	Severity	Mitigation
TAC replay failures for non-replayable operations	LOW	Audit application code for UTL_HTTP, DBMS_PIPE, NOCACHE sequences. Wrap in replayable transactions.
FastConnect provisioning lead time	MEDIUM	Order FastConnect in Phase 1. Use VPN as temporary connectivity until FC ready.
Team cloud skills gap	MEDIUM	Schedule OCI Foundations training. Engage Oracle Support for first 90 days.
Seasonal peak during migration window	HIGH	Complete migration before Black Friday. Maintain rollback capability for 30 days.

Well-Architected Scorecard

Pillar	Score	Status
Security & Compliance	20/22	✓ Pass
Reliability & Resilience	15/18	■ Pass with Recommendations
Performance & Cost	12/12	✓ Pass
Operational Efficiency	9/13	■ Pass with Recommendations
Distributed Cloud	—	— N/A

Top Recommendations

- Define quarterly DR drill schedule with automated validation
- Implement OS Management Hub for OKE node patching
- Configure Ops Insights for capacity planning
- Document operational runbooks before go-live

Validated against Oracle Well-Architected Framework — docs.oracle.com/en/solutions/oci-best-practices/

Next Steps

1. Review and approve architecture design — March 28, 2026
2. Provision OCI tenancy and order FastConnect — April 1, 2026
3. Run ADB assessment tool on production database — April 7, 2026
4. Begin Phase 1: Infrastructure setup — April 14, 2026
5. Schedule weekly architecture review cadence

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